



Massachusetts Institute of Technology

Freight Transportation Analysis

SCM.C51 Machine Learning in SCM

Today's Agenda

01

RESEARCH RECAP

Recap of the Interim presentation and guidelines for next steps

02

LIGHT GBM VS TRANSFORMERS INITIAL RESULTS

Deep-dive into exploratory data analysis, market behavior mapping, and seasonal decoding.

03

DEEP DIVE INTO EXOGENOUS DATA

Methodologies for data cleaning, demand construction, and exogenous market signal integration.

04

FEATURE ABLATION

Implementation of the LightGBM modeling framework and comprehensive analysis of initial results.

05

CONCLUSION

Transitioning insights into Temporal Fusion Transformers and multi-horizon forecasting strategies.

Research Recap
Recap of the Intern
presentation and
guidelines for next
steps

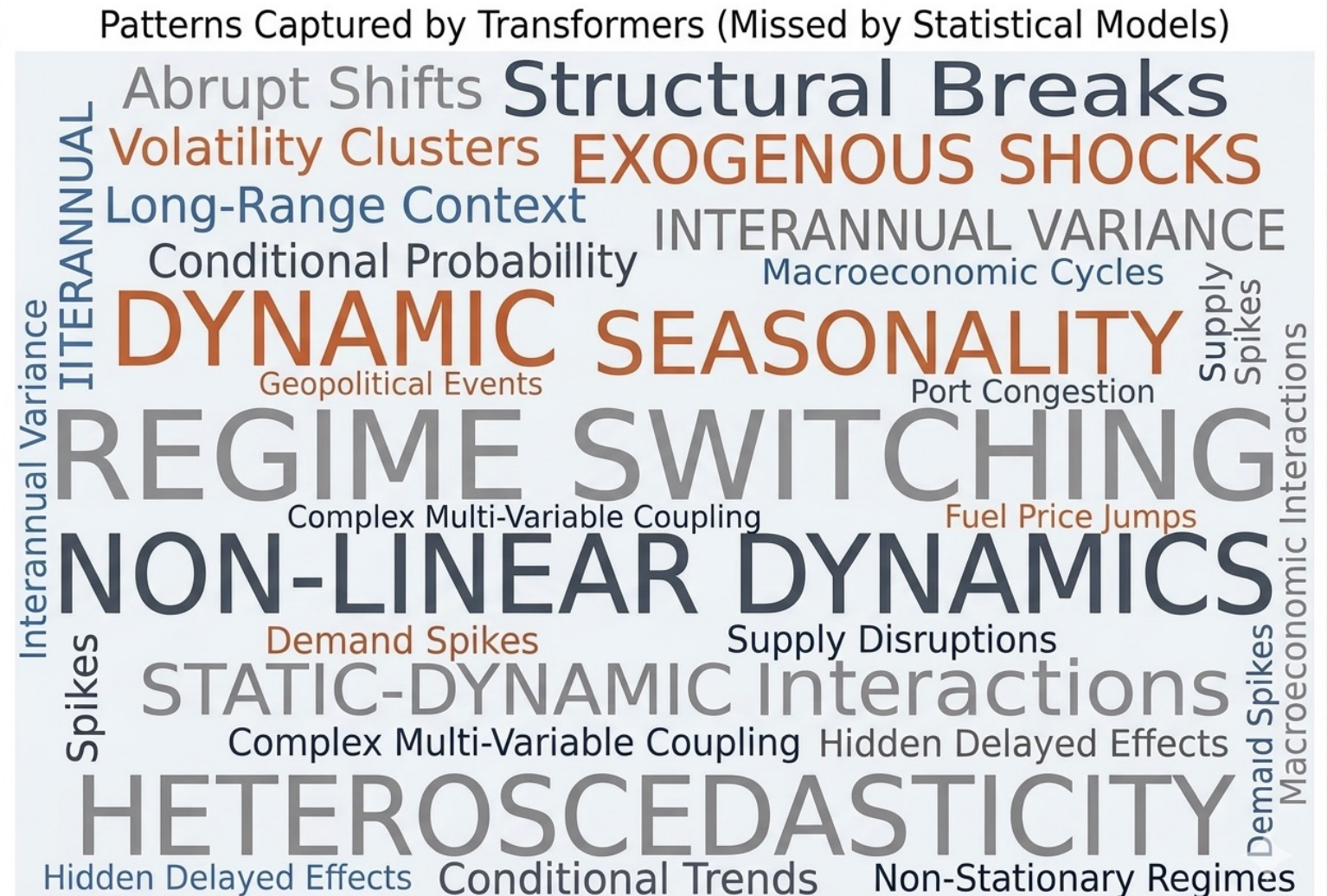
Can Transformer based models **capture nonlinear temporal dependencies and market dynamics** to improve lane-level freight rate forecasts compared to baseline models?

Our Research Question

Why is this important?

Freight pricing is spatially distributed and highly variable, reflecting dynamic, time-dependent behavior rather than simple static relationships.

Let's Deep-Dive
Into the Data

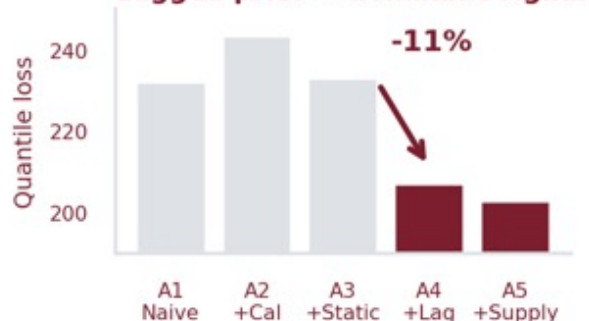


Why transformers for this problem?

Four properties of this problem favor attention-based sequence models

Memory is important

Lagged price = dominant signal



Lag features drive loss lower

Attention learns which past weeks matter, without hand-engineered lags.

Volatility and regime shifts

Wins the super-cycle fold

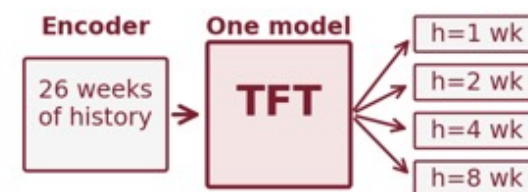


TFT could beat LightGBM in certain volatile market scenarios

TFT's longer-range memory could capture unprecedented spikes better than tree-based one-step lag dynamics.

Multi-horizon native

Multi-horizon in one pass

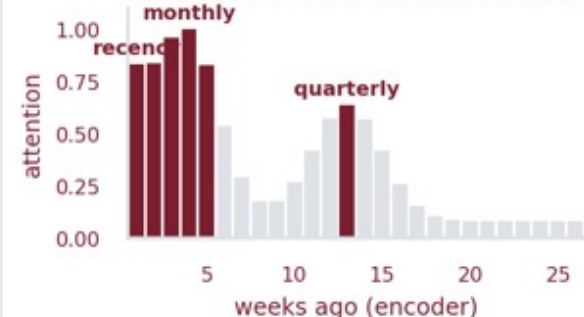


$h = 1, 2, 4, 8$ weeks in one pass

Single model, single training run, four horizons. No recursive one-step rollouts and no error accumulation - critical for $h = 8$ forecasts.

Interpretable by design

Attention shows what mattered

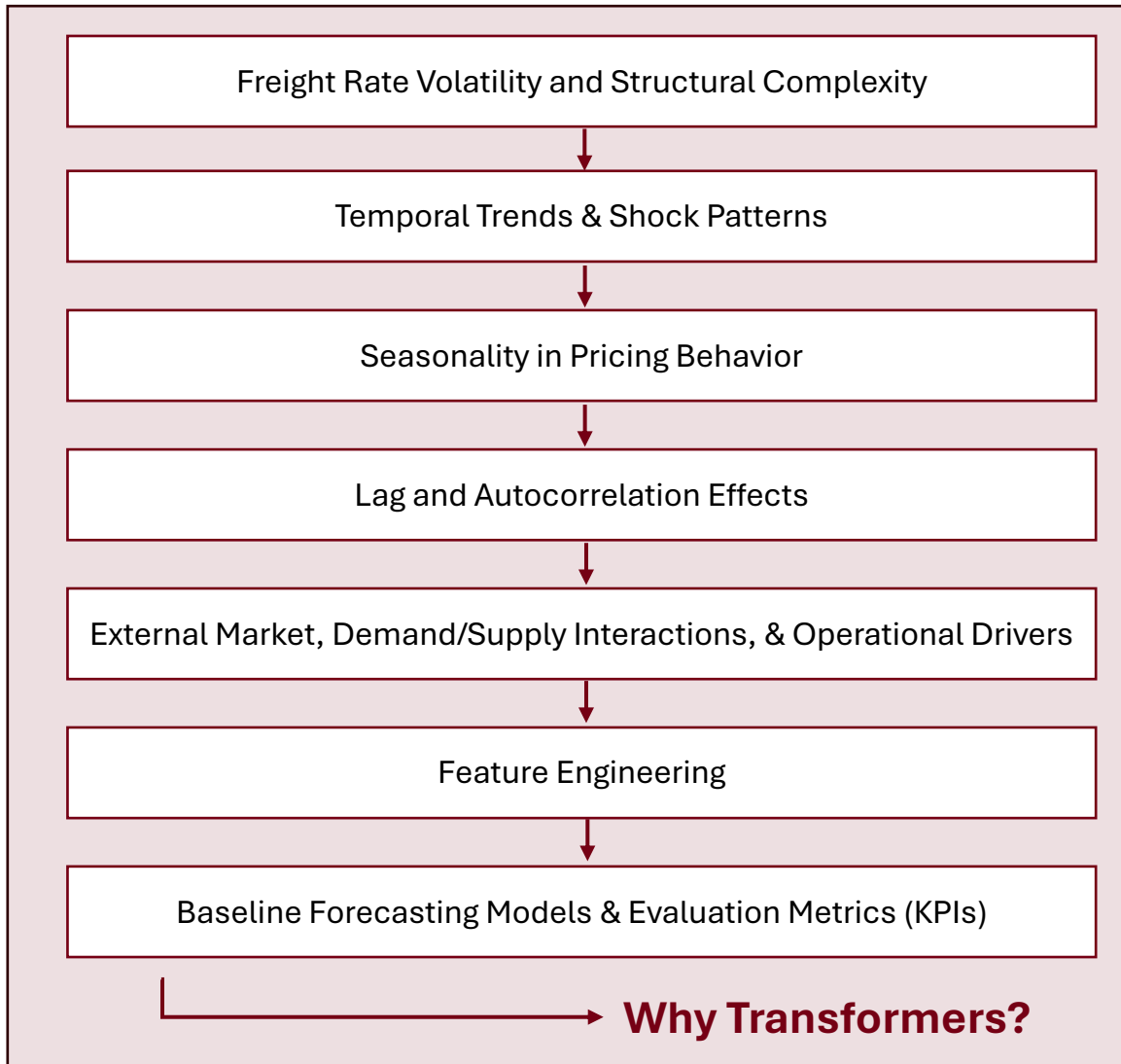


Attention + Variable Selection Networks

Per-prediction attention over past weeks and per-step variable selection weights expose WHICH features and WHICH weeks drove each forecast.

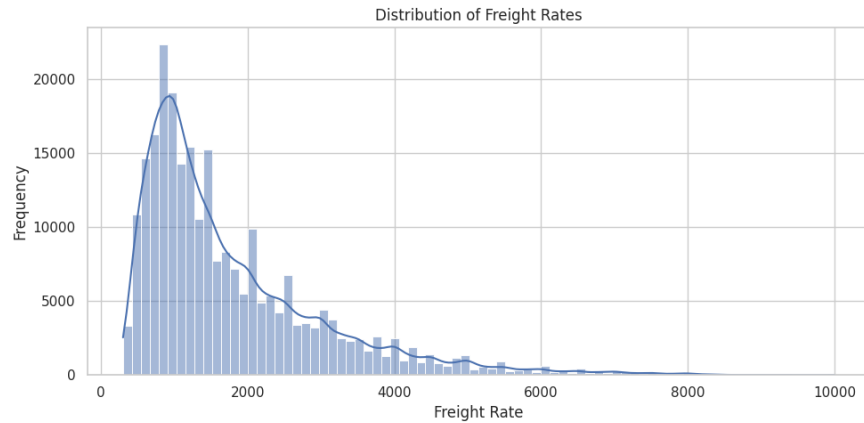
Scope Breakdown

How this Flows (Big Picture)

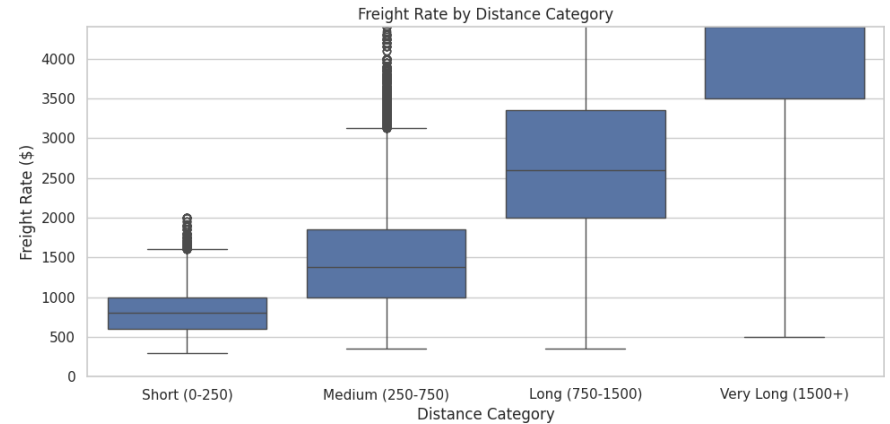


Market Behavior Overview

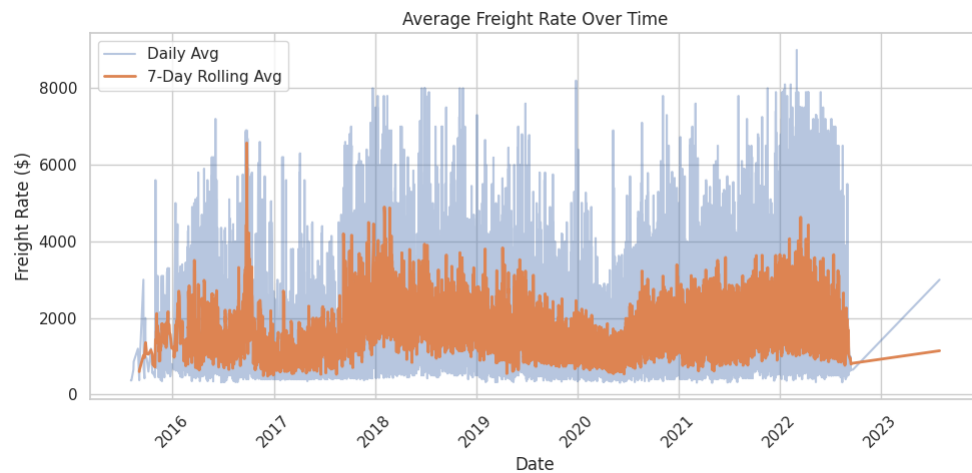
Freight pricing is spatially distributed and highly variable, reflecting dynamic, time-dependent behavior rather than simple static relationships.



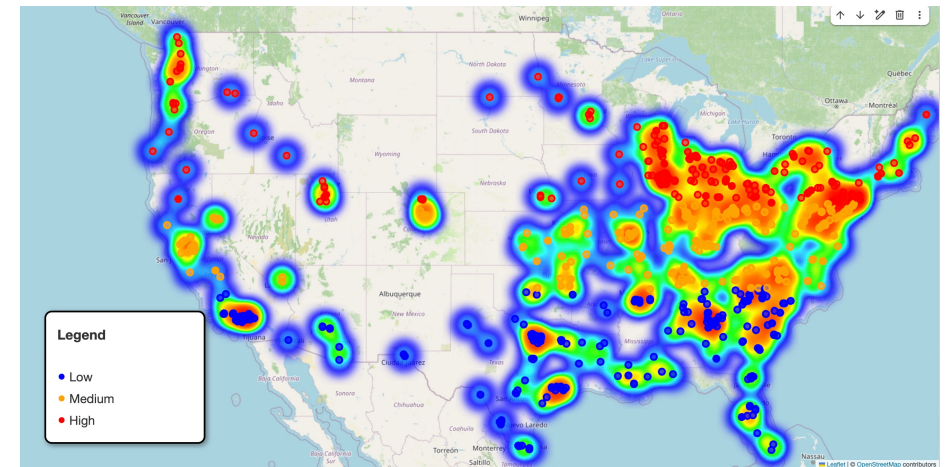
Freight rates are **highly dispersed and right-skewed**, with a range of pricing across shipments



Distance explains part of the pricing, but there is still **significant variation** within similar mileage ranges



Freight rates also **change over time**, indicating underlying temporal dynamics

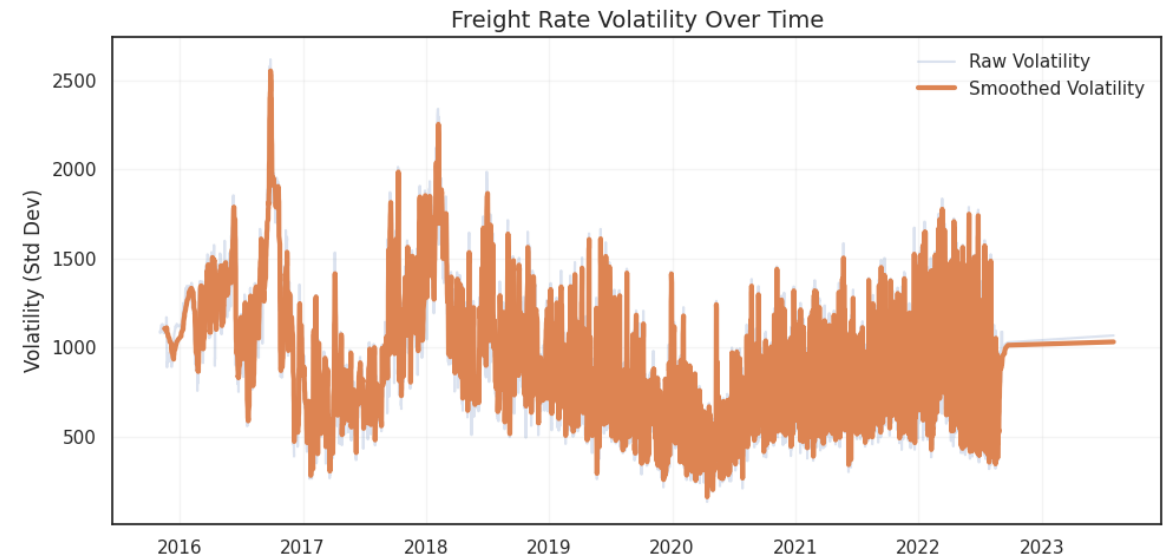
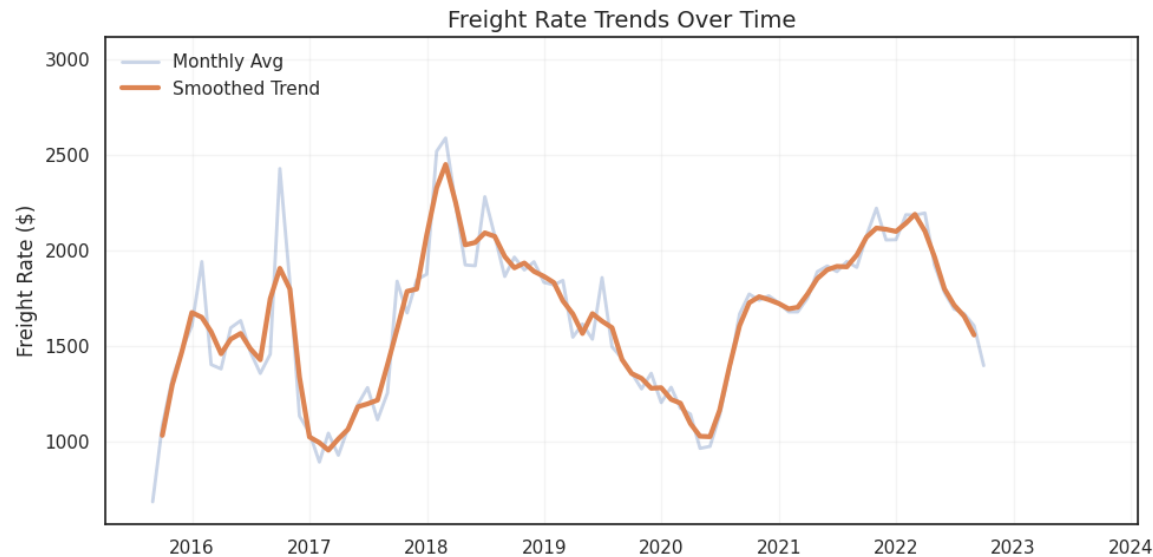


There is **clear geographic clustering** in prices, pointing to regional supply-demand imbalances

Temporal Dynamics: Trend and Structural Shifts

The time series shows non-stationarity in both mean and variance, violating assumptions of traditional statistical models

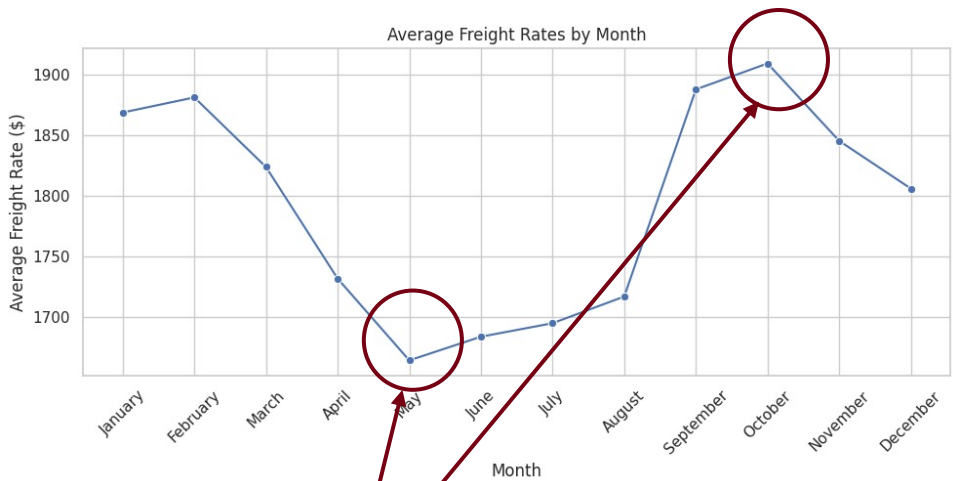
The freight market functions as a **dynamic, non-stationary system** characterized by **distinct regime shifts and time-varying volatility**,



confirming that pricing is driven by **complex, evolving market conditions** rather than fixed, predictable patterns.

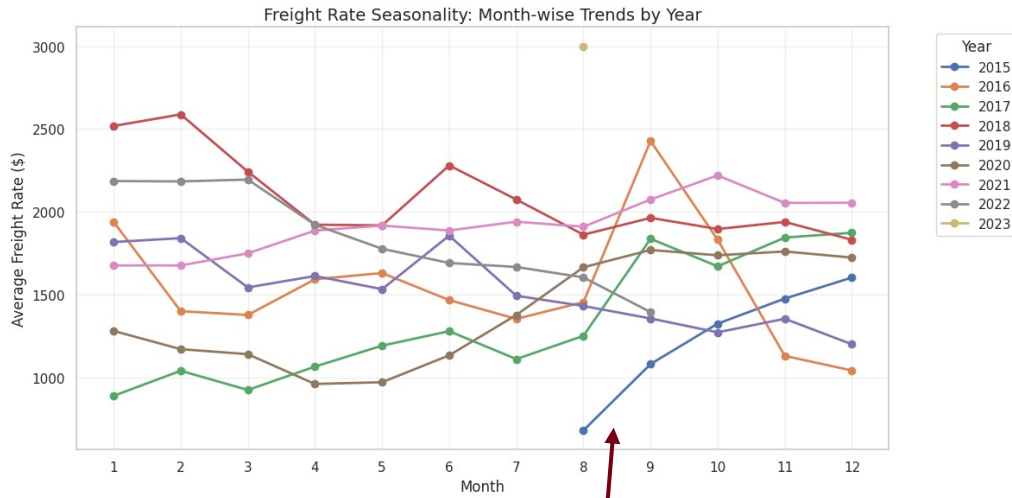
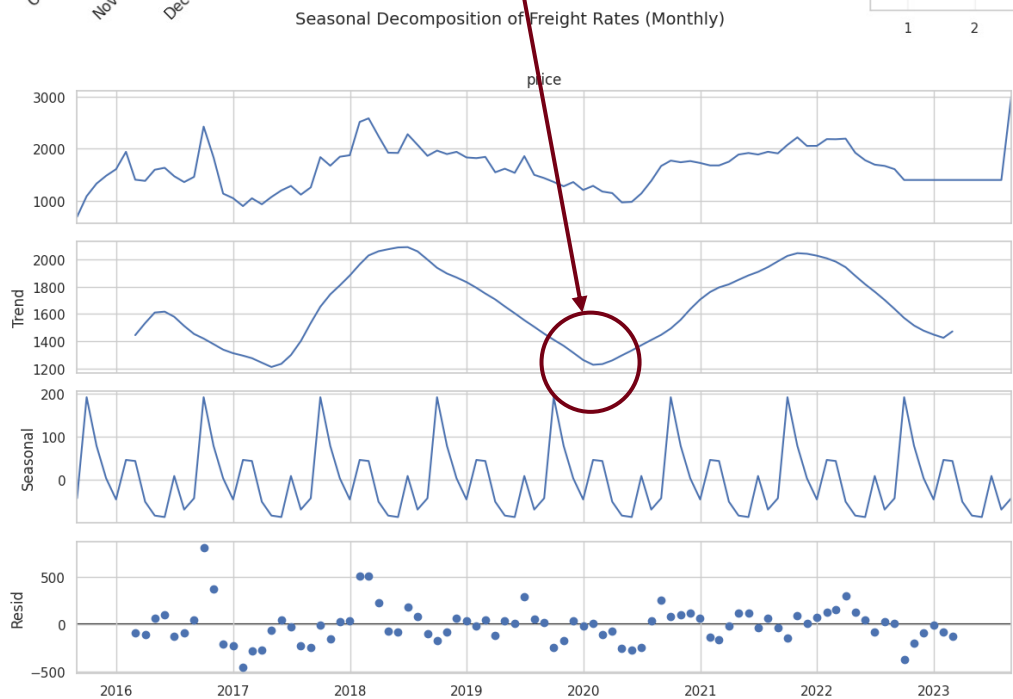
Decoding Freight Seasonality

Predictable cycles vs. structural market shifts in freight pricing



Seasonal decomposition reveals a **stable annual cycle** in freight rates, with recurring lows around and peaks around October

Trend seems to follow a certain economic cycle with disruption by **macroeconomic and supply demand shocks** clearly visible in 2020

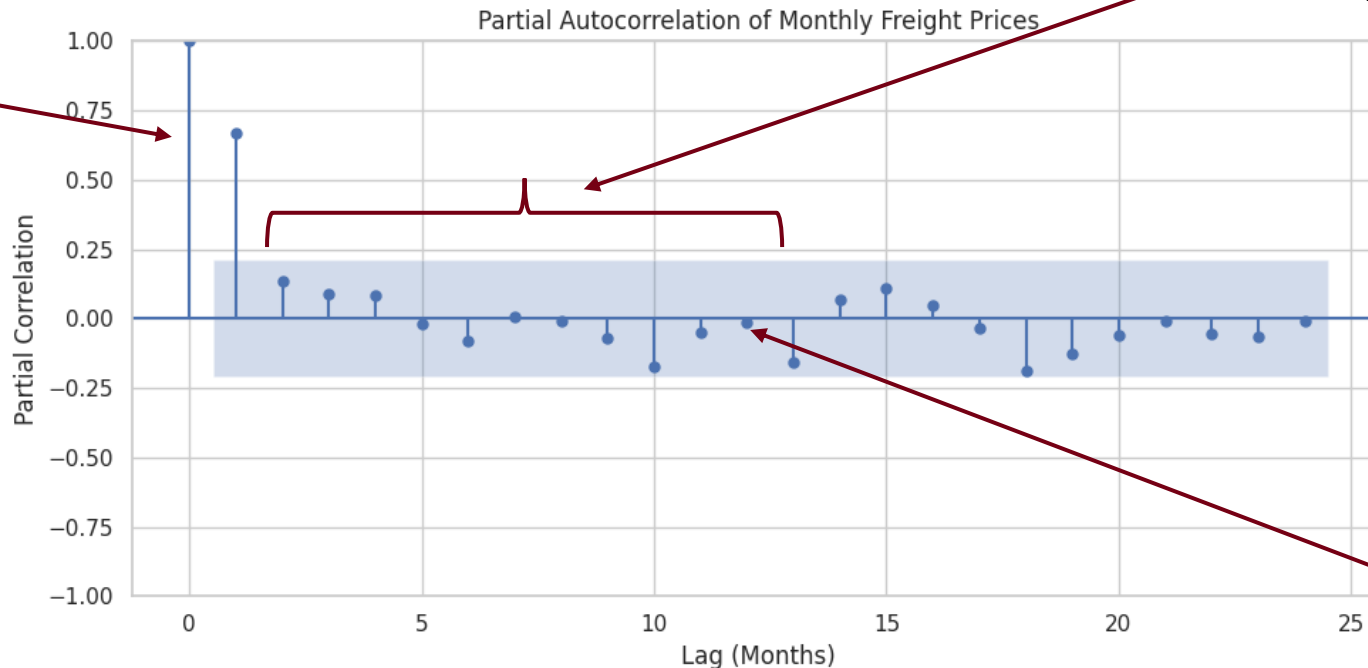


Residual variation remains significant and **seasonal behavior is not static**, suggesting the need for models that adapt to evolving temporal context rather than fixed assumptions

Autoregressive Structure in Freight Prices

Short-term persistence with rapidly decaying linear predictability

Strong spike at Lag 1 indicates **high short-term dependency** in freight prices, where recent values are the strongest predictors



Rapid decay in correlation after Lag 1 suggests **limited usefulness of linear relationships** for longer-horizon forecasting

Lack of a clear Lag 12 signal indicates that annual seasonality is not purely linear or directly repeatable year-over-year

The PACF confirms that while the market is highly path-dependent in the short term, its long-term drivers are non-linear and require a model capable of 'attending' to historical context without relying on direct linear correlations

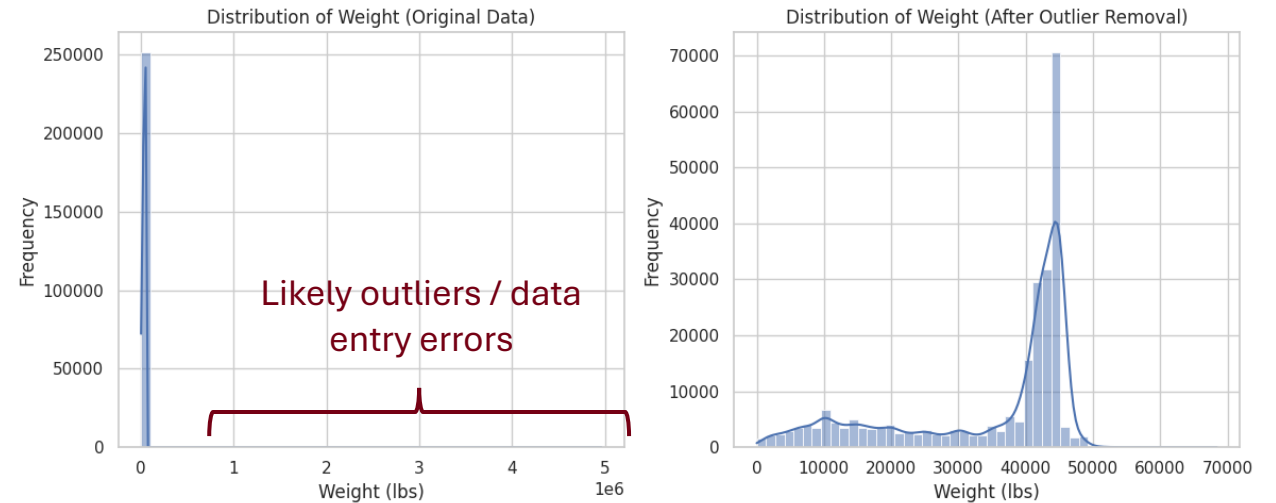
Freight rate dynamics involve **long-range temporal dependencies** and **interactions between multiple macroeconomic variables**. If sufficient high-frequency and multivariate data is available, transformer-based architectures may outperform tree-based models by learning cross-temporal and cross-feature attention patterns.

What the EDA tells us about potential modelling frameworks?

Data Cleaning & Preparation

Preparing freight data for modeling and feature engineering

After understanding the structure and behavior of the data, we performed **targeted cleaning** to ensure reliability for downstream modeling and feature engineering.



OUTLIER REMOVAL

- ✓ Removed unrealistic weights > 68,000 lbs
- ✓ Addressed entry errors up to ~6M lbs
- ✓ Applied robust boxplot-based filtering
- ✓ Result: Industry norm ~40k lbs avg



FEATURE CONSISTENCY

- ✓ Detected `team_driver` flag mismatches
- ✓ Resolved 963 conflicting records
- ✓ Ensured feature set (X) reliability
- ✓ Created `team_driver_X` & `_base`



TIME MISALIGNMENT

- ✓ Standardized `hour_of_day` & `pickup_date`
- ✓ Resolved encoding inconsistencies
- ✓ Ensured temporal consistency for modeling
- ✓ Separated time (`hour`) and date fields

While only 36 rows were removed, cleaning ensures that model inputs reflect true operational behavior rather than data artifacts, improving both accuracy and interpretability. Additionally, it brings to light potential

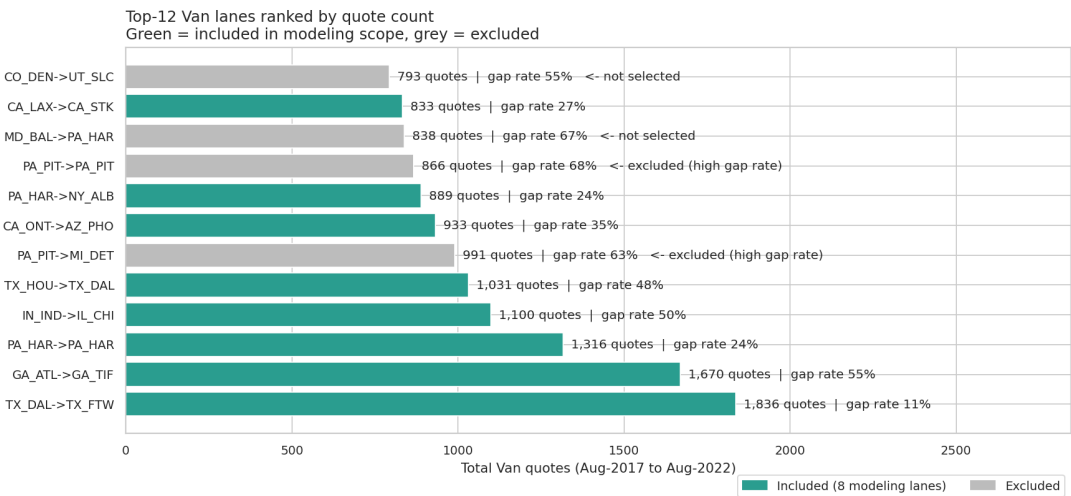
Lane-Level Freight Forecasting Framework

Transitioning from exploratory analysis to setting data up for modelling

Equipment: Van Only. EDA highlighted that equipment type is a key price driver. We restricted to van quotes (220,795 rows, ~88%) to avoid conflicting pricing regimes

Top Van Lanes: Ranked by total quote count to provide sufficient panel observations for robust time-series modeling.

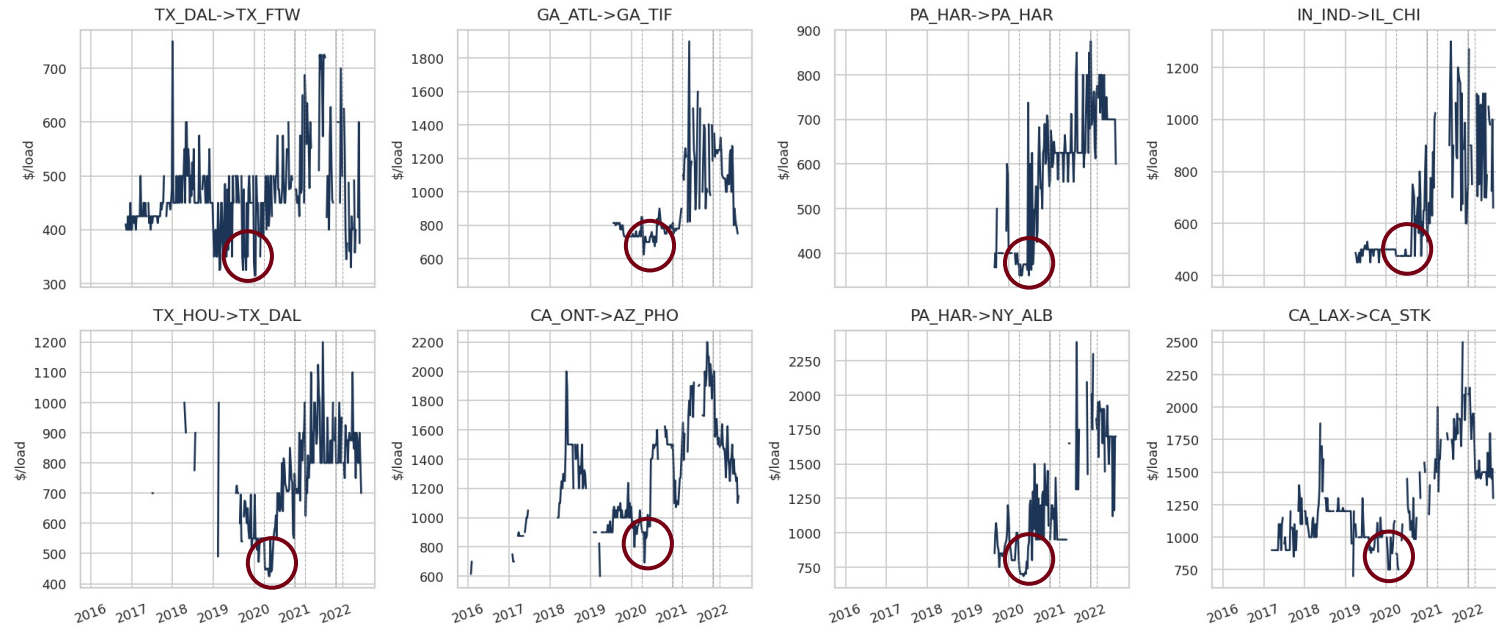
Gap Rate Threshold: Dropped any lane with a weekly gap rate above ~50%. Sparse lanes break sequence models and inflate training noise.



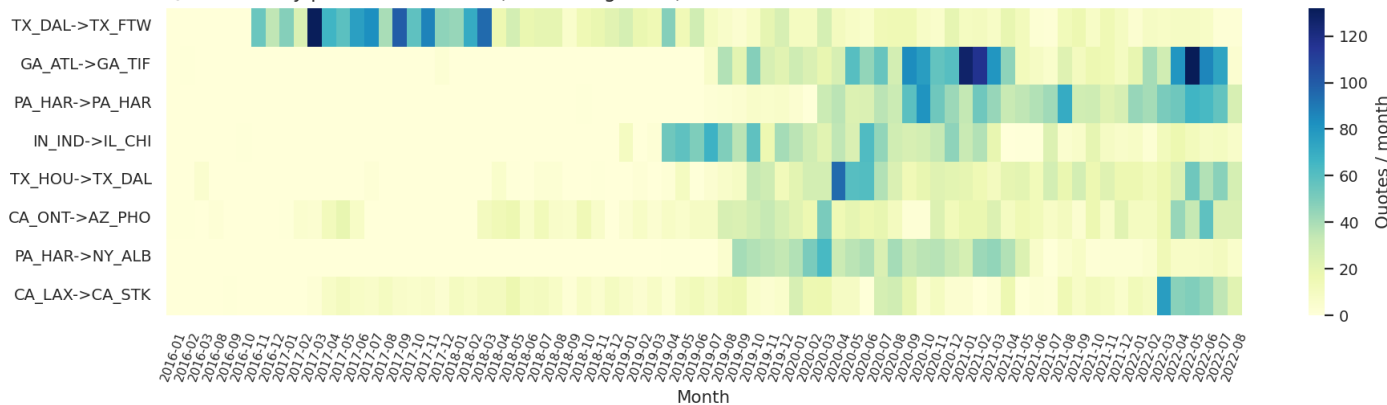
LANE (ORIGIN -> DEST)	TOTAL QUOTES	WEEKS ACTIVE	GAP RATE	MEDIAN PRICE	MEDIAN MILEAGE	QUOTES/WEEK
TX_DAL -> TX_FTW	1,836	302	11%	\$450	131 mi	3
GA_ATL -> GA_TIF	1,670	339	55%	\$814	250 mi	0
PA_HAR -> PA_HAR	1,316	201	24%	\$625	61 mi	6
IN_IND -> IL_CHI	1,100	305	50%	\$552	214 mi	1
TX_HOU -> TX_DAL	1,031	337	48%	\$750	261 mi	1
CA_ONT -> AZ_PHO	933	343	35%	\$1,250	316 mi	2
PA_HAR -> NY_ALB	889	176	24%	\$1,078	287 mi	4
CA_LAX -> CA_STK	833	308	27%	\$1,200	329 mi	2

Lane (KMA) Level Price Dynamics & Activity

Freight Markets Exhibit Regime Shifts, Volatility, and Uneven Activity Across Lanes



Quote density per lane over time (8 modeling lanes)



What's happening in lane-level data?

- Freight rates exhibit **strong non-stationarity**, with a clear 2021 super-cycle spike followed by uneven normalization across lanes
- Cross-lane consistency in the shock period indicates a **systemic market regime shift rather than lane-specific noise**
- Volatility and trend structure vary significantly by lane, reflecting **heterogeneous pricing dynamics across corridors**
- Quote density reveals **highly uneven and time-varying market activity**, reinforcing demand-driven pricing behavior
- Emergence and sparsity patterns across lanes highlight that freight markets are **irregular, evolving, and structurally non-uniform over time**

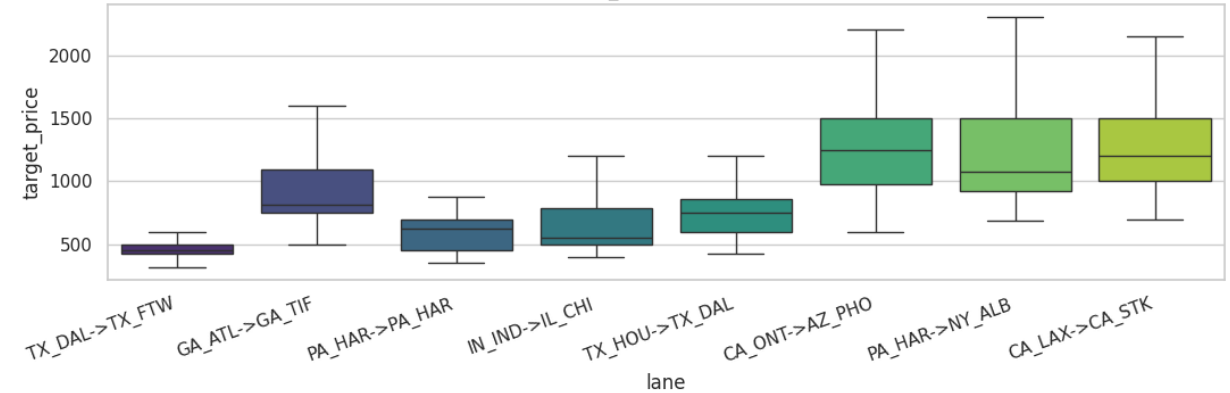
Lane Level Structure & Freight Price Formation

How routing (Origin to Destination corridors) shapes price dynamics, stability, and market heterogeneity

Weekly quote volume by lane



Weekly target_price distribution by lane



What's happening in lane-level data?

- Lane-level structure clearly plays a meaningful role in shaping target prices, suggesting that **routing (O-D corridor effects) is an important pricing driver** beyond just temporal variation
- Grouping by lane helps **reduce noise from cross-corridor heterogeneity** and results in more stable, more interpretable price patterns over time
- Quote volume differs substantially across lanes, indicating persistent imbalance in market activity and liquidity across different corridors
- There are clear **distributional differences between lanes**, with some showing tighter price bands and others exhibiting higher dispersion and volatility

What are we predicting? Duration of Freight Price for Prediction

Restricting analysis to high-volume corridors for stable temporal learning

THE CORE OBJECTIVE

Predicting the future weekly median freight price for specific lanes, including uncertainty bounds (P10–P90), across multiple time horizons.



THE TARGET

Weekly Median Price of all freight quotes per lane.
Robust to outliers and representative of the market level.



UNCERTAINTY

Probabilistic distribution via Quantile Regression:
P10 (Cheap), P50 (Likely), and P90 (Expensive) scenarios.

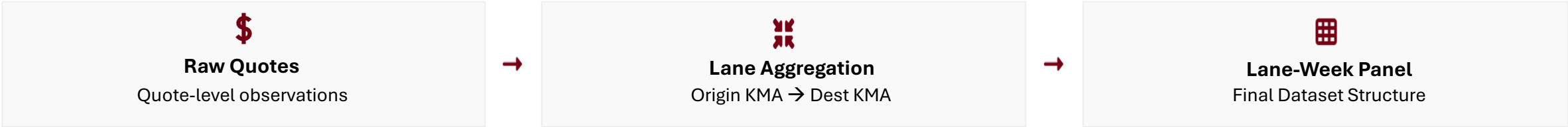


HORIZONS

Multi-horizon forecasting at 1, 2, 4, and 8 weeks ahead to support operational, tactical, and strategic planning.

Feature Engineering at Lane (KMA) Level

Transforming raw quote-level data into structured time-series inputs for forecasting models (56 total modelling features selected)



lane	week	horizon	price	lag_1	lag_4	roll_4_mean	quote_count	matched_carriers	mean_weight	week_sin	month_cos	mileage	origin_lat	dest_lat
TX_DAL→TX_FTW	2021-W10	2	500	480	460	470	7	12.5	39,500	0.87	0.50	131	32.8	32.7
CA_ONT→AZ_PHO	2021-W10	4	1250	1200	1100	1150	3	9.2	42,100	0.87	0.50	316	34.0	33.4

Each row represents one (lane × week) observation with engineered features

Identifiers

lane, horizon: Informs the model exactly where the lane is and how far ahead it needs to forecast.

Lagged & Rolling

price_lag_X, price_lag_4, price_roll_*:
Captures market memory and autoregressive dynamics.

Supply & Demand

quote_count, matched_carriers_mean:
Real-time proxy for market tightness and pricing pressure.

Weekly Aggregates

price, mean_price, std_price: Captures the full distribution and dispersion of prices within a week.

Calendar Encoding

week_sin, month_cos: Encodes cyclical seasonality without assuming fixed, rigid patterns.

Static Lane Features

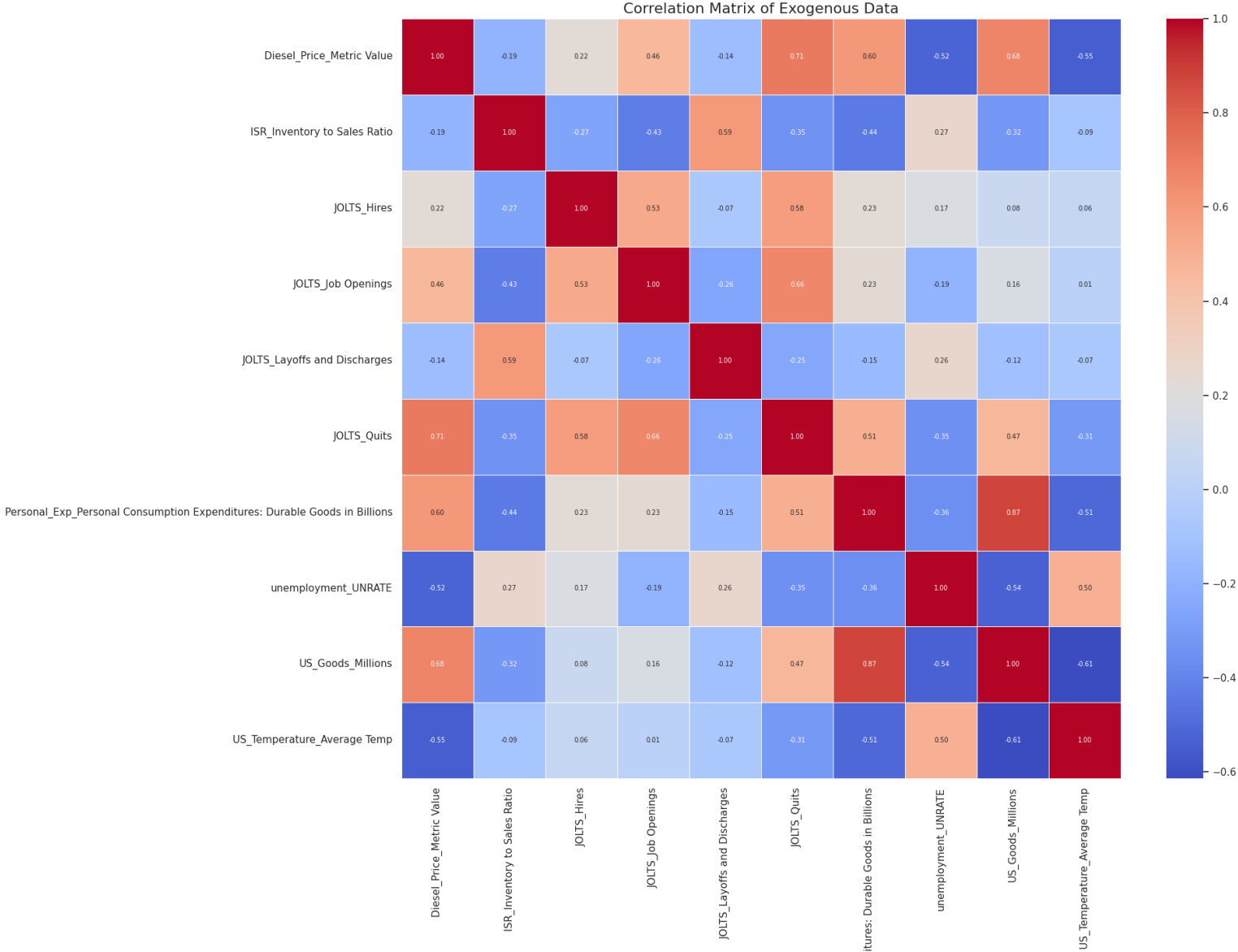
mileage, lat/long, urbanicity: Encodes fixed structural differences across the network geography.

Incorporating Exogenous Market Signals

Freight pricing is not purely internal: it reacts to macroeconomic conditions

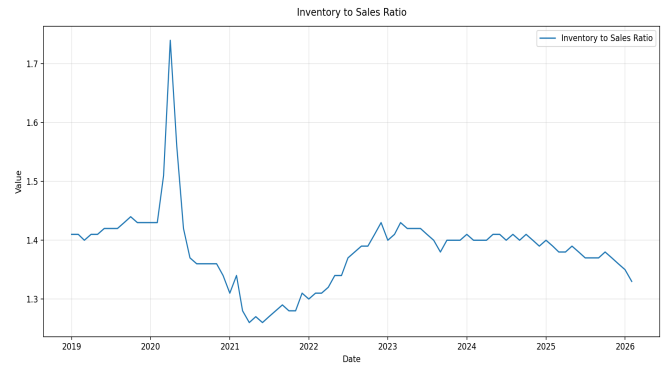
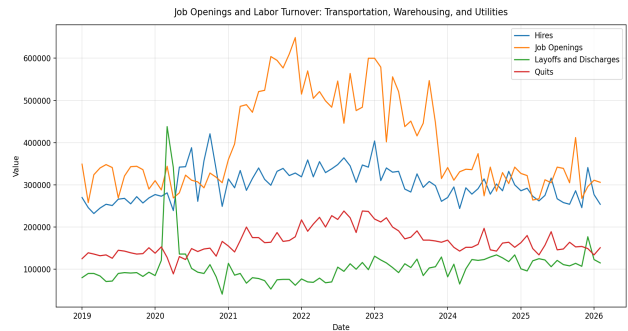
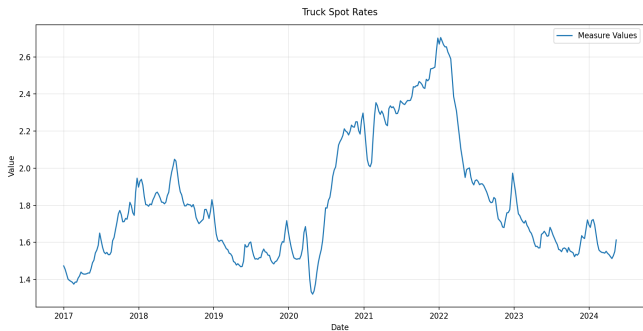
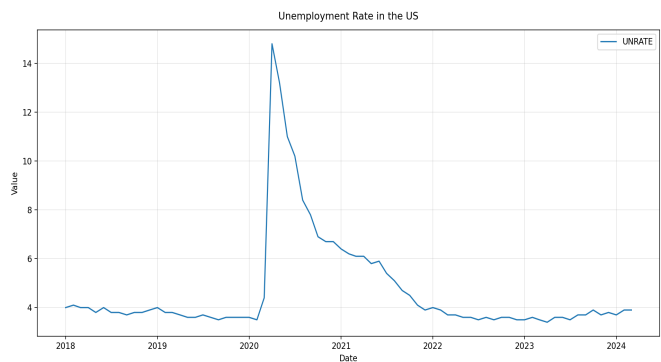
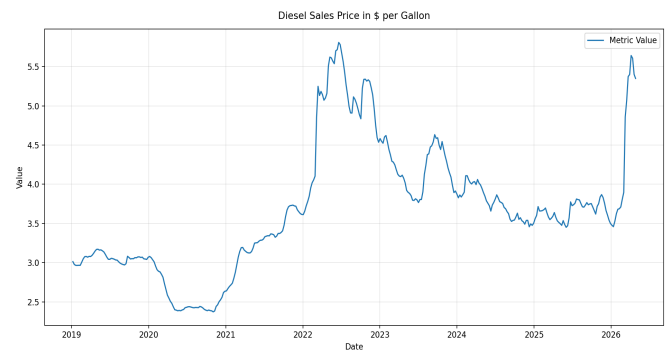
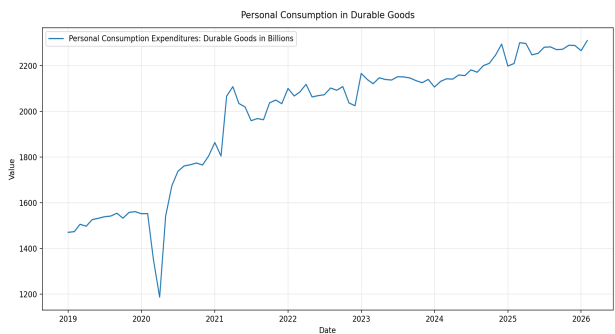
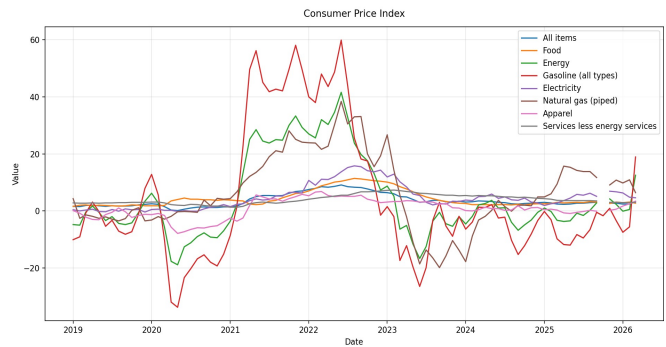
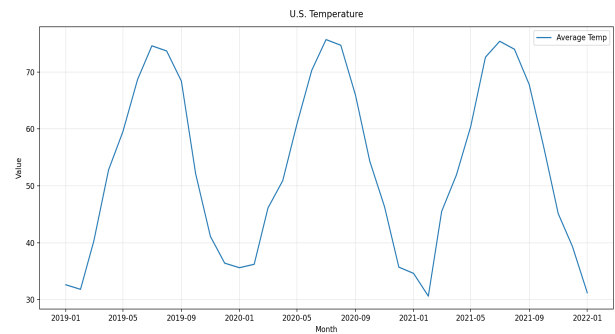
Current Exogenous Features

- Consumer Price Index
- Diesel Sales Price in \$ per Gallon
- Inventory to Sales Ratio
- Job Openings and Labor
- Turnover: Transportation, Warehousing, and Utilities Sector
- Personal Consumption in Durable Goods in \$ Billions
- Truck Spot Rates in \$ per mile by truck type
- Unemployment Rate in the US
- US Goods Imports in Millions
- US Temperature



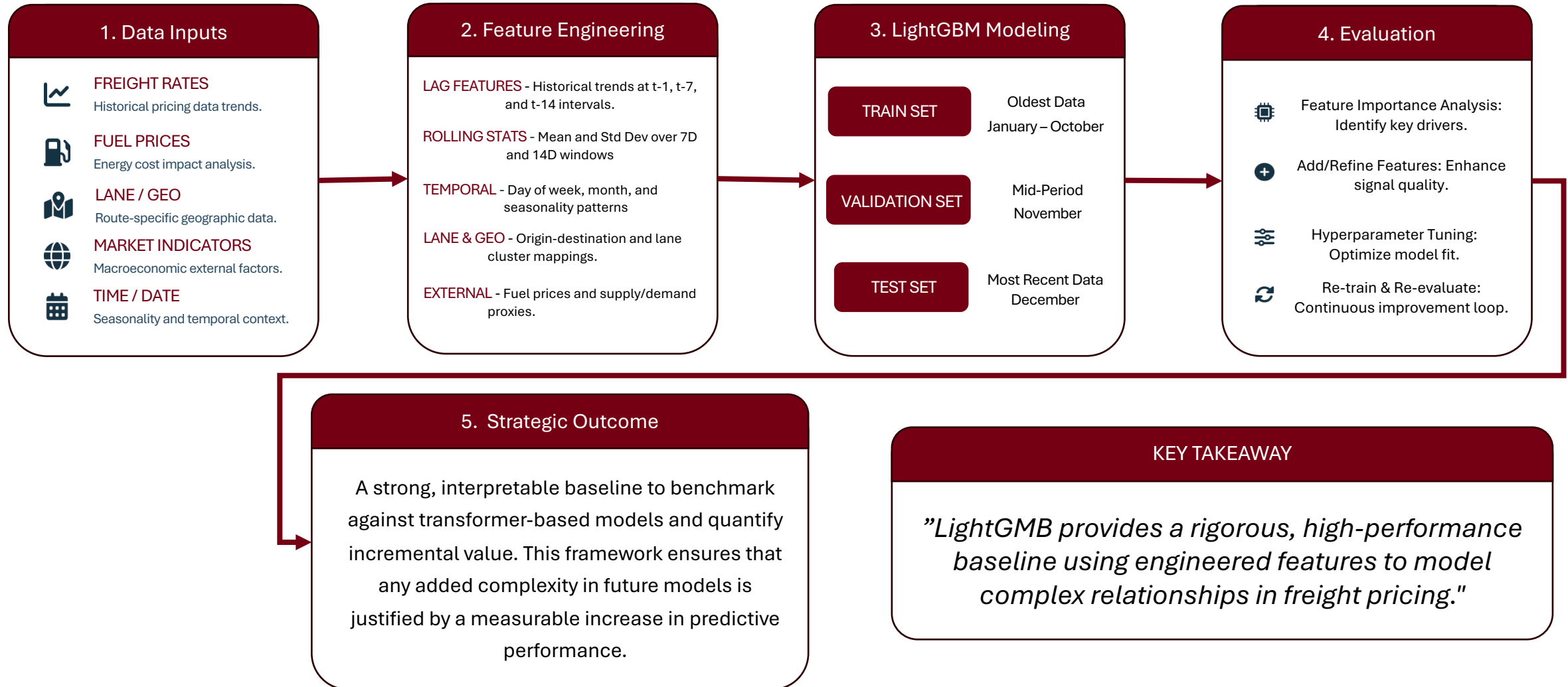
Incorporating Exogenous Market Signals

Freight pricing is not purely internal: it reacts to macroeconomic conditions



Baseline Modeling Framework

Not all external variables contribute equally to pricing dynamics

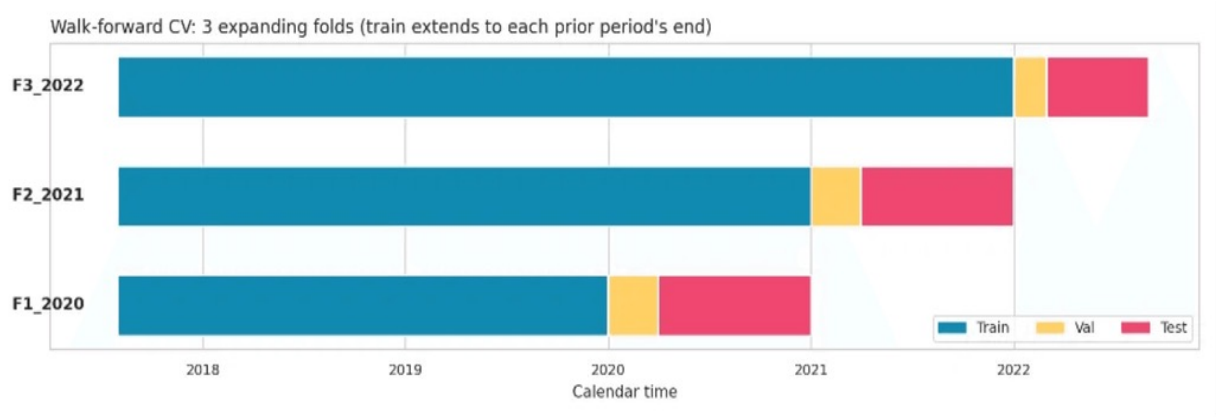


LightGBM Modeling Strategy

Using Walk-Forward Validation & Framework Architecture

WHY THIS APPROACH?

- ✔ **Non-Stationary Market Dynamics:** Freight markets undergo structural shifts (e.g., COVID supply chain shocks). Standard validation fails to account for these regime changes, leading to poor generalization.
- ✔ **Temporal Realism & Leakage:** In production, we only have historical data. Walk-forward CV strictly respects the time arrow, ensuring the model never "sees" the future during training or validation.
- ✔ **Expanding Window Memory:** By expanding the training set over time, we allow the model to learn both long-term seasonal priors and recent market volatility across multiple economic cycles.



FOLD	TRAINING PERIOD	VALIDATION (EARLY-STOPPING)	TEST PERIOD (OUT-OF-TIME)	ROWS
F1_2020	2017-08 ... 2019-12-31	2020-01-01 ... 2020-03-31	2020-04-01 ... 2020-12-31	1,134
F2_2021	2017-08 ... 2020-12-31	2021-01-01 ... 2021-03-31	2021-04-01 ... 2021-12-31	966
F3_2022	2017-08 ... 2021-12-31	2022-01-01 ... 2022-02-28	2022-03-01 ... 2022-08-31	612

FRAMEWORK: LIGHTGBM

A high-performance Gradient-Boosted Decision Tree (GBDT) model. It can capture complex, non-linear interactions in tabular time-series data while maintaining computational speed.

MODELING VARIANTS

- v1: Raw price baseline (Default)
- v2: Log-transformed target (Stability)
- v3: Tuned log model (Optimized)
- v4: CQR calibration (Uncertainty)

EVALUATION METRICS

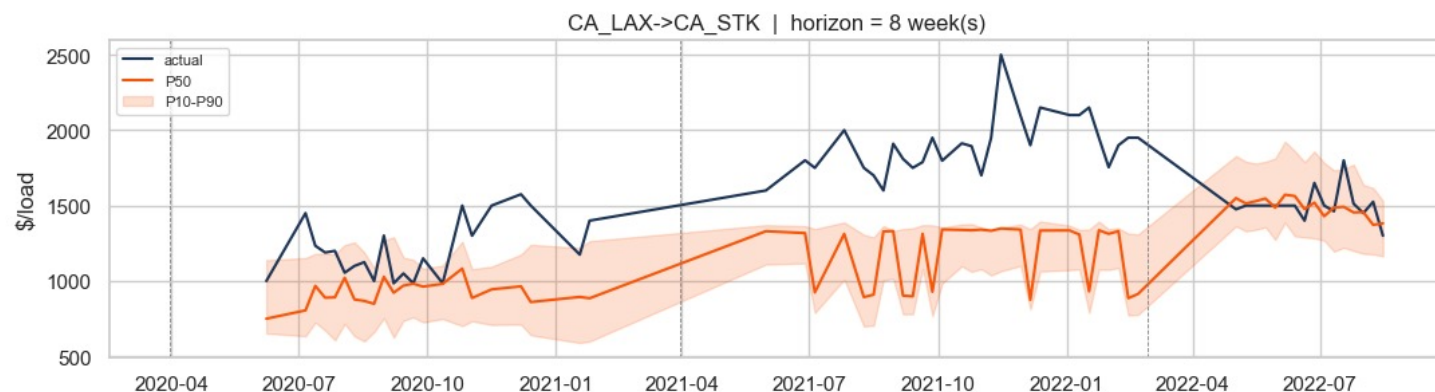
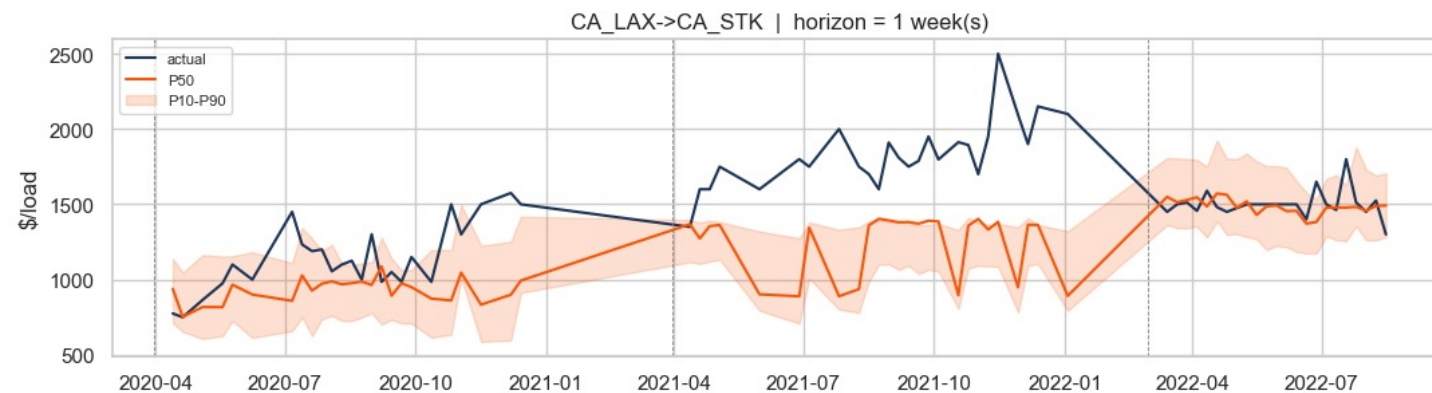
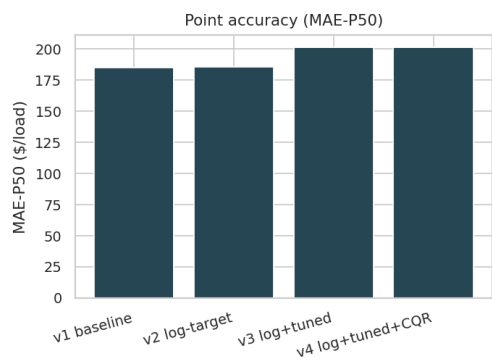
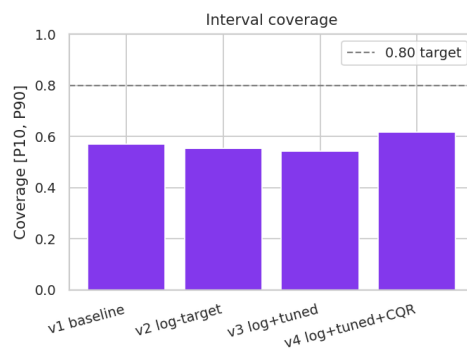
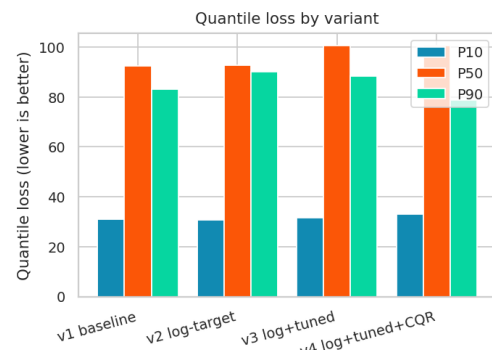
Multi-dimensional performance tracking:

MAE MAPE Quantile Loss Coverage

Focusing on both point accuracy and interval reliability.

LightGBM Modeling Strategy

Using Walk-Forward Validation & Framework Architecture



LightGBM shows the best overall performance with the lowest error and strong short-term predictions. However, it underestimates uncertainty and struggles during sharp market shifts, with accuracy decreasing at longer horizons.

LightGBM Results

Not all external variables contribute equally to pricing dynamics

 **FEATURE DOMINANCE**

~ 85%

OF TOTAL IMPROVEMENT

Lagged Price Features

Supply Signals (2021 Shift)



Calendar features alone increased error by 5% as 2021 broke seasonal priors.

 **REGIME SENSITIVITY (MAE)**



NORMAL MARKET (2020)

\$160 MAE

High Predictability



SUPER-CYCLE (2021)

\$276 MAE

Fundamental Accuracy Ceiling

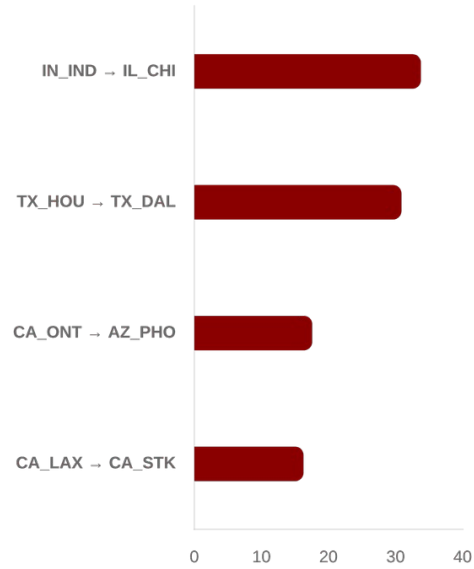


COOL-DOWN (2022)

\$131 MAE

Optimal Performance

 **LANE-LEVEL GAINS (%)**



Corridor	Gain (%)
IN_IND → IL_CHI	~32
TX_HOU → TX_DAL	~30
CA_ONT → AZ_PHO	~18
CA_LAX → CA_STK	~15

*Comparison of A1 (Naive) vs A5 (Full Feature Set). Volatile long-haul corridors benefit most.

LightGBM emerges a strong baseline, consistently leveraging lagged price structure and engineered features to deliver stable, well-calibrated forecasts across lanes and horizons, while more advanced sequence-based approaches are a natural next step for capturing longer-range dependencies and potential regime shifts that may not be fully expressed in the current feature set.

Why is LightGBM selected as the primary forecasting model, and what future modeling directions could further improve performance in this problem?

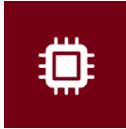
Next Steps: From Insights to Transformer Based Forecasting

Transitioning from exploratory analysis and baseline modeling to advanced modeling



WHAT WE HAVE ESTABLISHED

- Complex temporal dynamics identified (**Trend, Seasonality, AR**)
- Feature engineering captures demand signals and temporal structure
- Exogenous variables introduce macroeconomic market context
- Baseline models establish performance benchmarks and KPIs



NEXT STEP: TRANSFORMER MODELING

- Integrate Feature Ablation to LightGBM and Transformers
- Add Exogenous Features to “improve” LightGBM problem
- Train a **Temporal Fusion Transformer (TFT)** on lane-level time series
- Integrate temporal, demand-driven, and exogenous features
- Multi-horizon forecasting: **1, 2, 4, 8 weeks**
- Benchmark performance against baseline models



EXPECTED OUTPUTS & INSIGHTS

- Point forecasts (**P50**) for actionable pricing decisions
- Probabilistic forecasts (**P10–P90**) for uncertainty quantification
- Feature importance via attention and variable selection
- Insights into dynamic relationships across time and features

Move from static-feature based traditional time series models to Transformers that learn dynamic interactions and uncertainty in freight pricing